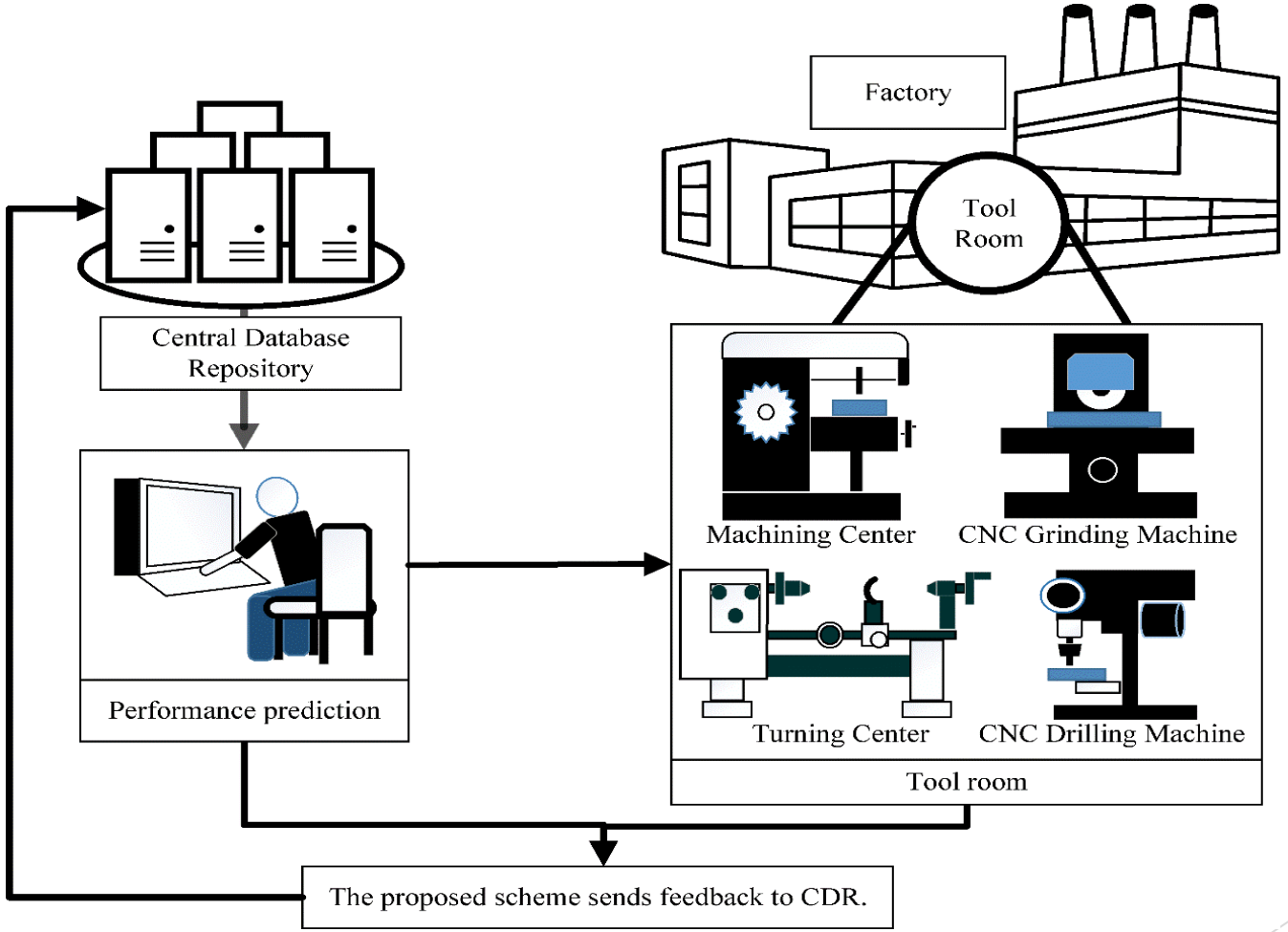
BREAK-EVEN ANALYSIS, Part-1

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Two type of costs

- Fixed Costs (F_c) (like building rent, salaries)
- Variable Costs (V_c) (material cost)



Courtesy: Ph.D. Thesis of Kaustabh Chatterjee

What is Contribution?

Contribution=Sales – Variable costs

Contribution – Fixed Costs=Profit

Loss=Negative of Proft

Profit= Sales – Variable costs– Fixed Costs

Suppose fixed cost is Rs. 1000/-. Variable cost per unit is Rs. 100/-.

Cost of making one unit=Rs. 1000/-+Rs. 100/-=Rs. 1100/-

Cost of making two units=Rs. 1000/-+Rs. 100/-$\times$2=Rs. 1200/-

$\Rightarrow$Per unit cost = Rs. 1200/2=Rs. 600/-

Selling price is decided by market.

Profit or Loss

- Suppose Selling Price=Rs. 200/-
- If only one unit is made, then

Contribution=Sales–Variable Cost =200 –100=Rs. 100/-

Profit=Contribution –Fixed Cost=100 –1000=(–)Rs. 900

Thus, we make a loss of Rs. 900/-.

If two units are made, then

Contribution=Sales–Variable Cost =200×2 –100×2=Rs. 200/-

Loss=Fixed Cost – Contribution =1000 –200=Rs. 800

Loss per unit=Rs. 800/2=Rs. 400/-

Break-Even Point

- At break-even point, contribution will be equal to fixed cost.
- There is no profit and no loss.
- Business will make loss if amount of contribution is lower than the amount of fixed cost.
- Business will make profit if amount of contribution is greater than the amount of fixed cost.
- Contribution < Fixed Cost $\Rightarrow$Loss
- Contribution > Fixed Cost $\Rightarrow$Profit
- Contribution = Fixed Cost $\Rightarrow$Break-Even
- Break-Even point=Fixed Cost / (Contribution per unit)

An example

Q.: The rent of a school building is Rs. 30000/- per month. Expense on teachers' salary is Rs. 70000/- per month. On each student Rs. 500/- per month is spent on mid-day meal. School fee for a student is Rs. 1000/- per month. How many students should be admitted to reach break-even point?

Solution: Total fixed cost=Rs. 30000/-+Rs. 70000/- =Rs. 100,000/-

Contribution per unit= Sales (fee) per unit –Variable cost per unit

=1000 – 500=Rs. 500

Break-Even point=Fixed Cost/Contribution per unit=100,000/500=200 students

Profit calculation

- Profit=Total Sales–Fixed cost – total variable cost
- =Total fee collection–Fixed cost – total expense on mid-day meal
- $=1000\times200 -100{,}000 -500\times200=0$
- Thus, we see that at Break-Even Point, the profit is zero.
- Contribution is called by this name because it is the contribution of each unit sold towards the organization's bottom line (profit).
- If an organization is making losses, then with every increase in quantity sold will reduce loss to the extent of contribution.

Another example

- Sales per unit is Rs. 300. Variable cost is Rs. 100 per unit. Fixed cost is Rs. 10000. Suppose 100 units are sold. We can divide fixed cost into 100 units. Hence, fixed cost per unit=10000/100=Rs. 100.
- Profit per unit =300−100−100=Rs. 100
- Suppose one additional unit is sold at Rs. 100. It may appear that there will be loss of (100+100 − 100)=Rs. 100.
- But this calculation is not correct.
- Let us calculate it correctly.
- When 100 units are sold at Rs. 300 per unit. Total Sales=300×100=30000. Fixed Cost=10000, Variable cost=100×100=10000
- Total Profit = 30000−10000−10000=Rs.10000
- When additional unit is sold at Rs. 100. Total Sales=300×100+100=30100. Fixed Cost=10000, Variable cost=100×101=10100
- Total Profit = 30100−10000−10100=Rs.10000
- Contribution of additional unit=Sales −variable cost=100−100=0.
- Hence, it neither increases nor decreases the profit.

Importance of Contribution

- If contribution is positive, the sale would be profitable.
- If organization is making profit, then for additional sales, the profit will increase to the extent of contribution.
- If it is suffering losses, losses will reduce to the extent of contribution.

$$\text{Break-Even Point} = \frac{\text{Fixed Cost}}{\text{Contribution per unit}}$$

Profit-Volume Ratio

$$\text{PV Ratio} = \frac{\text{Contribution}}{\text{Sales}};$$

$$\text{Contribution} = \text{Sales} \times \text{PV Ratio}$$

An example

- Selling price of one unit=Rs. 300, Variable cost of one unit=Rs. 150.
- Contribution per unit= Selling price –variable cost=300 – 150=Rs. 150
- PV Ratio=(contribution/sales)=(150/300)=0.5 or 50%.
- Suppose 100 units are sold. Total sales value=300×100= Rs. 30000.
- Contribution=Rs. 30000 × PV Ratio= Rs. 30000 × 0.5=Rs. 15000.
- Let Fixed Cost be Rs. 10000.
- Profit=Contribution – Fixed cost=15000–10000=Rs. 5000

Important Equations

- Profit=(Sales ×PV Ratio)–Fixed Cost
  - Contribution =Fixed Cost+ Profit

$$\text{Sales}=\frac{\text{Contribution}}{\text{PVRatio}}$$

$$\text{Sales}=\frac{\text{Desired Profit+Fixed Cost}}{\text{PV Ratio}}$$

An Example

- The selling price per component is Rs. 50. The variable cost per component is Rs. 10. Fixed cost per day is Rs. 10000. Find out the break-even point.

Solution:

$$\text{Break-even point} = \frac{\text{Fixed cost}}{\text{Contribution per unit}};$$

$$\text{Contribution per unit} = \text{Selling price per unit} - \text{varaible cost per component}$$

$$= 50 - 10 = \text{Rs.}40;$$

$$\text{Break-even point} = \frac{10000}{40} = 250 \text{ components per day}$$

Profit by selling 300 components

- Total sales=50×300=Rs. 15000.
- Contribution=15000–10×300=Rs. 12000
- Profit =Contribution –Fixed Cost =12000-10000=2000 per day.

Profit by Selling 600 components

- Total sales=50×600=Rs. 30000.
- Contribution=30000–10×600=Rs. 24000
- Profit =Contribution –Fixed Cost
  =24000–10000=14000 per day.
- Thus. doubling the production does not double profit, but increases it by 7 times.
- This called leverage effect.

Case: Desired Profit Rs. 15000 per day

- Profit =Rs. 15000
- Fixed Cost =Rs. 10000
- Contribution =Profit + Fixed cost =15000+10000=Rs. 25000

$$\text{PVRatio} = \frac{\text{Contribution}}{\text{Sales}} = \frac{40}{50} = 0.8$$

$$\text{Sales} = \frac{\text{Contribution}}{\text{PVRatio}} = \frac{25000}{0.8} = 31250\,\text{units}$$